

Impact Assessment: AIIA Clinical Trials Dashboard

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1. Purpose of this document

This document assesses the social, economic and environmental impact of the **AIIA Clinical Trials Dashboard**, a cloud-based, GCP-compliant Clinical Trial Management System (CTMS) for Ayurveda research. It complements the research paper draft (Method, findings, limits) by focusing specifically on *who benefits, in what way, and by how much*.

The system under assessment carries an Ayurveda study from protocol through IEC approval, CTRI registration, enrolment, scheduled visits, adverse-event (AE/SAE) capture and final reporting, with nine role-based views. Regulatory rules (GCP-ASU, NDCT Rules 2019, CTRI status, pharmacovigilance timelines) are encoded as executable rules rather than manual checklists, and every change is written to an immutable, time-stamped audit trail aligned to ALCOA+ principles.

2. Baseline: the problem being addressed

Ayurveda clinical research in India is still conducted on spreadsheets and email. Three failure modes follow, and each maps to a distinct impact pathway:

Failure mode	Consequence today
Safety data is chased, not captured	AE/SAE signals surface late; reporting clocks are missed
Status reports are recompiled by hand	~2 h/week per portfolio of coordination time
Provenance is reconstructed after the fact	Audits are a reconstruction exercise, not a query

Regulatory exposure is material: GCP-ASU, the CTRI registry, the NDCT Rules 2019 and the DPDP Act 2023 all impose obligations that are currently met through human diligence rather than software guarantees.

3. Social impact

3.1 Benefit to research participants

- Participants gain direct visibility of their own study record, visit schedule and ID card through a portal designed to be readable by a non-specialist — an explicit design goal, since existing trial paperwork is not written for patients.
- Consent status becomes visible and enforceable in software rather than held in a file, reducing the risk of a participant being enrolled or dosed without valid, current consent.
- Equity of access improves for participants at sites with slow connectivity, because visit capture is designed mobile-first rather than desktop-bound.

3.2 Benefit to clinicians and study staff

- Investigators and coordinators replace multiple spreadsheets with one record; enrolment, visits and queries appear in a single place.
- Ethics committees receive approval-expiry and deviation signals *before* they become findings at audit, enabling action in days rather than months.
- Senior clinicians face a lower adoption barrier because the dashboards are designed as zero-training surfaces, with shadowing sessions during onboarding.

3.3 Benefit to pharmacovigilance and regulators

- AE and SAE records are routed on regulatory timelines rather than chased over email, protecting participants from late-identified safety signals.
- NPvCC and pharmacovigilance centres gain a reconciled view of open safety cases.
- Regulators and auditors receive a complete, immutable ALCOA+ trail instead of reconstructed files, raising the evidential quality of Indian Ayurveda research.
- Public confidence in Ayurveda research improves as provenance becomes demonstrable rather than asserted.

3.4 Benefit to the wider research ecosystem

- Because CTRI, EDC and hospital information systems expose no usable public APIs, CSV and FHIR R4 bulk import is the realistic integration path. Interoperability built on FHIR R4 and CDISC means Indian Ayurveda data becomes structurally comparable with global health data, improving the evidence base for ASU therapies.

4. Economic impact

4.1 Cost displacement

The dominant economic effect is replacing a large one-off licence with a small recurring operating cost:

Indicator	Today (manual)	With the dashboard	Change
Software cost	₹10–15 lakh (~\$12k–18k)	₹2,500–6,000 / \$30–70 a month	–95%
Status reporting time	~2 h / week, by hand	Live, ~15 min / week	–87%
Audit trail	Reconstructed after the fact	Immutable, time-stamped	Always on

These figures are **modelled from a stated manual baseline**, not measured in a production deployment. They are presented here as feasibility evidence and must be validated in a live 12-study portfolio before being cited as measured results.

4.2 Impact on institutional budgets

For an AYUSH college or research institute, the shift converts a capital-style procurement decision into an operating line item that can be absorbed by existing annual budgets, and removes a procurement bottleneck that often blocks trial-management software entirely.

4.3 Impact on the wider AYUSH economy

Lower compliance-recording cost per trial raises the expected return on running an Ayurveda trial, which should increase trial throughput. Increased trial volume generates additional demand for CROs, IECs, NPvCC services and data management — a multiplier effect across the regulatory ecosystem.

5. Environmental impact

5.1 Paper reduction

A zero-print dashboard replaces most of the paper previously printed per study each month:

Indicator	Today (manual)	With the dashboard	Change
Paper per study / month	40+ printed pages	~6 pages	–85%

Only genuinely required records — signed consent originals and regulatory dossiers — would remain on paper, which is both environmentally preferable and regulatorily appropriate.

5.2 Serverless resource efficiency

The deployment targets a serverless edge runtime (Cloudflare Workers, D1, R2) rather than always-on virtual machines. The running prototype reports a ~10 ms cold start and a ~31 KB payload. Because compute is consumed only while serving a request, idle infrastructure consumes no energy — a materially better profile than a permanently provisioned trial-management server, particularly for institutions with intermittent demand.

5.3 Reduced travel and courier traffic

Replacing emailed attachments and couriered consent documentation with a single live record reduces inter-site document movement. This effect is directional and unquantified at present.

6. Impact indicators to be tracked

Adoption is the dominant open risk in this project, and the only open risk budgeted rather than assumed. The following indicators should be tracked during any pilot:

Dimension	Indicator	Target direction
Social	Time for an IEC to act on a deviation	Days, not months
Social	Participant records self-accessed	Rising
Social	Staff adoption (active users vs. named roles)	Rising
Economic	Status-reporting hours per week	–87%

Economic	Software cost per month	–95% vs. licence
Environmental	Printed pages per study per month	–85%
Environmental	Idle compute energy	Near zero (serverless)

7. Limitations of this assessment

1. **No live clinical deployment.** No impact figure has been observed in an operational AYUSH trial; all figures are modelled from the stated manual baseline.
2. **Single-institution extrapolation.** Baseline assumptions (2 h/week, 40+ pages) reflect one observed workflow pattern and may not hold at larger multi-site institutions.
3. **Adoption is unproven.** Social benefits are conditional on clinicians changing working practice. Zero-training design and shadowing are mitigation strategies, not evidence.
4. **Environmental figures exclude hardware.** The assessment covers paper and compute idle-time only; participant travel and device manufacture are not counted.
5. **No independent verification.** Nothing in this document has been externally audited or peer reviewed.
6. **Participant-facing benefit is designed, not validated.** The participant portal has not been tested with real participants or assessed for comprehension.

8. Conclusion

The AIIA Clinical Trials Dashboard is expected to produce measurable social benefit for participants, clinicians, ethics committees and regulators; substantial economic benefit through a ~95% reduction in software cost and ~87% reduction in reporting effort; and environmental benefit through an ~85% reduction in printed pages alongside near-zero idle compute. All quantified figures are modelled rather than measured, and adoption remains the single largest open risk — which is why the indicators in Section 6 are specified for tracking rather than assumed.

9. References

- Clinical Trials Registry – India (CTRI) — ctri.nic.in
- Ayush Suraksha (pharmacovigilance) — suraksha.ayush.gov.in
- GCP / ICMR-National Ethical Guidelines (CDSR) — cdsr.nic.in
- NDCT Rules 2019 — indiacode.nic.in
- DPDP Act 2023 (MeitY) — meity.gov.in
- HL7 FHIR R4 — hl7.org/fhir
- CDISC — cdisc.org
- All India Institute of Ayurveda, Ministry of Ayush — aiia.gov.in

Status: assessment for review. Quantified values are modelled from a stated manual baseline and require pilot validation.